ADITYA PENUMARTI

aditya-penumarti.github.io
linkedin.com/in/adi-pen • Google Scholar

RESEARCH INTERESTS

Robot decision-making under uncertainty in unstructured and GPS-denied environments: belief-space planning, uncertain multimodal representations, information-constrained planning, and active perception for marine and ground robots in GPS-denied environments.

EDUCATION

Ph.D. Mechanical Engineering Expected Spring 2027
University of Florida, Gainesville, FL
Dissertation: *Uncertainty Quantification and Representations for Active Perception in Navigation and Planning Frameworks*
Advisor: Jaejeong Shin (now Seoul National University)
Committee Chair: Rushikesh Kamalapurkar

M.S. Mechanical Engineering August 2023
University of Florida, Gainesville, FL
Thesis: *Mechanical Networks: An Investigation Into Connected Kinematic Systems*
Advisor: Carl D. Crane III

B.S. Mechanical Engineering May 2021
North Carolina State University, Raleigh, NC

PUBLICATIONS

†Student mentee

Journal Articles

[J1] **Aditya Penumarti**, Humberto Ramos, Kristy Waters, Kevin Brink, Jaejeong Shin, "Real-Time Uncertainty-Aware Motion Planning for Magnetic-Based Navigation," *IEEE Access*, vol. 14, pp. 23824–23834, 2026, doi: 10.1109/ACCESS.2026.3661848.

[J2] Jaekwang Kim, Hyung-Jun Park, **Aditya Penumarti**, Jaejeong Shin, "Fast Marching Based Rendezvous Path Planning for a Team of Heterogeneous Vehicles," *IEEE Access*, vol. 12, pp. 122609–122623, 2024, doi: 10.1109/ACCESS.2024.3444314.

Conference Papers

[C1] **Aditya Penumarti**, Khanh Dong†, Zi-Hao Zhang†, Yongkyoon Park, Zhenqi Wu, Trung Dong, Shahriar Negahdaripour, Xiaomin Lin, Jaejeong Shin, "Calibration and Comparative Analysis of Forward-Looking Sonar and 3D Sonar for Enhanced Underwater Object Recognition," *IEEE/MTS OCEANS 2026 - Monterey*, 2026.

[C2] Trung Tien Dong, Zhenqi Wu, **Aditya Penumarti**, Zi-Hao Zhang†, Micaiah Bartlett, Jaejeong Shin, Xiaomin Lin, "uScenes: A Multimodal RGB and 3D Sonar Dataset for Underwater Robot Perception," *IEEE/MTS OCEANS 2026 - Monterey*, 2026.

[C3] Zi-Hao Zhang†, E. Baker Herrin, Jia Guo, **Aditya Penumarti**, Zilong He, Andres Pulido, Jaejeong Shin, "Action Recognition for Underwater Gesture Communication in Human Diver and Robot Teaming," *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.

[C4] Gokul Puthumanaillam, **Aditya Penumarti**, Manav Vora, Paulo Padrao, Jose Fuentes, Leonardo Bobadilla, Jaejeong Shin, Melkior Ornik, "Belief-Conditioned One-Step Diffusion: Real-Time Trajectory Planning with Just-Enough Sensing," *Proceedings of the 9th Conference on Robot Learning (CoRL)*, PMLR 305:68–92, 2025.

[C5] E. Baker Herrin, **Aditya Penumarti**, Andres Pulido, Nikhil Iyer†, You-Long Wu†, Kaleb Lewellen, Tyler Fitzsimmons, Jaejeong Shin, "Lessons from RAITE: Real-World Evaluation of Robust Multimodal Target Detection and Tracking under Adversarial Attacks," *Disruptive Technologies in Information Sciences IX, Proc. SPIE*, 2025.

[C6] **Aditya Penumarti**, Nikhil Iyer†, Loi Park†, Kristy Waters, Jose H. Ramos, Jaejeong Shin, "Neural Network-Based Magnetometer Sensor Model for Virtual Magnetic-Based Navigation Testbed," *AIAA SciTech Forum*, 2025.

Under Review

[R1] **Aditya Penumarti**, Nikhil Iyer†, Jaejeong Shin, "Bayesian Expandable Ambient Magnetic Mapping (BEAMM): Continual Learning with a Physics-Constrained Bayesian Neural Network," submitted to *IEEE Robotics and Automation Letters (RA-L)*, 2026.

[R2] **Aditya Penumarti**, Jaejeong Shin, "HALo: Hilbert-space Active Localizability for Scalar Field Navigation," submitted to *2027 IEEE International Conference on Robotics and Automation (ICRA)*, 2027.

[R3] Manav Vora, Gokul Puthumanaillam, **Aditya Penumarti**, Jaejeong Shin, Melkior Ornik "Akule: Fast and Scalable Multi-Robot Motion Planning via Sparse Interaction Diffusion" submitted to *2027 International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2027.

RESEARCH EXPERIENCE

University of Florida, Gainesville, FL: Graduate Research Assistant August 2023 - Present
Active perception and uncertainty representations in GPS-denied environments (Advisor: Jane (Jaejeong) Shin)

- Developed an uncertainty-aware magnetic navigation planner that, compared with state-of-the-art methods, improves localization accuracy by 5%, cuts failure rates by 34%, and reduces computational costs by over 99% [J1].
- Built neural and Bayesian neural network models of magnetometer sensing and magnetic fields, including expandable maps with competitive calibration to Gaussian process baselines [C6, R1].
- Characterized how gradients in scalar fields determine localizability and used this for information-driven planning [R2].
- Designed research drones for information-aware guidance; built a magnetic navigation simulator in NVIDIA Isaac Sim, Gazebo, and ROS2. [C6]
- Developed low- and high-level software for an underwater vehicle (thrust mapping, filtering, closed-loop control) and a gesture-driven diver-following controller with transformer-based gesture recognition [C3].
- Built radar–vision fusion tracking (YOLO detection, global nearest-neighbor association) for the U.S. Navy NSWC Crane Robust AI Test Event (RAITE), evaluated under adversarial attack [C5].

Center for Intelligent Machines and Robots, UF, Gainesville, FL: Lab Assistant September 2021 – August 2023
Research on mechanically connected systems to evaluate kinematic feasibility (Advisor: Carl D. Crane III)

- Automated a 1/10th vehicle platform with GNSS and LiDAR navigation; implemented ROS2 path planning, SLAM, LiDAR mapping and localization, and UKF-based sensor fusion.
- M.S. thesis (Advisor: Carl D. Crane III): mechanical network analysis of closed-loop mechanisms (four-bar, Stephenson, and parallel) for geometric mobility, joint displacement, and internal joint preload.

TEACHING EXPERIENCE

Course Co-Developer, EML4842C: Autonomous Vehicles, University of Florida Fall 2022

- Co-developed the course with Carl D. Crane III
- Designed and maintained robotic vehicle platforms, created assignments, projects, and lecture materials in conjunction with the lecturer.
- Taught undergraduate students autonomous navigation using GNSS and LiDAR on a 1/10th autonomous vehicle platform.

Teaching Assistant, University of Florida 2021-2023, 2025

- EML2023 - Computer Aided Design, Fall 2021, Summer 2025, Graded assignments, and held office hours
- EML6282 - Robot Geometry II, Spring 2022, Graded assignments and held office hours.
- EML3301C - Mechanics of Materials Lab, Summer 2025, graded assignments and provided office hours for students

UF-MAE Rapid Prototyping Facility, Gainesville, FL: Manager and Lab TA September 2021 – August 2023

- Trained and managed student assistants in CNC machining and 3D printing to precise tolerances.
- Designed and built apparatus and experiments for undergraduate laboratory courses: (1) Computer Aided Design, (2) Mechanics of Materials Lab, (3) Dynamics and Controls System Laboratory

Guest Lectures

- *Lecture on "Uncertainty-Aware Motion Planning for Magnetic-based Navigation,"*, EML6934 Spring 2026 [J1]
- *Lecture on Dead Reckoning*, EML4842C: Autonomous Vehicles, Spring 2023
- *Lecture on Particle Filtering*, EML6934: Sensor-Based Robot Planning - Spring 2024
- *Lecture on Controllability*, EML4312: Control of Dynamic Systems - Fall 2024

MENTORING

- Zi-Hao Zhang, master's and Ph.D. researcher (2024-Present)
  Learning-based tracking of divers and agents [C1,C2,C3]
- Khanh Dong, master's researcher (2026-Present)
  Computer vision and sonar calibration [C1]
- Nikhil Iyer, undergraduate researcher (2023-2026)
  Magnetometer modeling and mapping [C5,C6,R1]; *now a software engineer at AutoPCB*
- You-Long Wu, master's researcher (2024-2026)
  Multi-modal fusion [C5]; *Now a lecturer at the National Defense University, Taiwan*
- Loi Park, undergraduate researcher (2024-2025)
  Magnetometer modeling [C6]; *Now a Senior in mechanical engineering at the University of Florida*

PROPOSAL WRITING EXPERIENCE

- "A Unified Risk-Aware Representation for Multi-Modal Planning in Uncertain Environments," Amazon Research Awards 2026
  Role: Developed the contributions, methodology, and experimental plan. PI: Jaejeong Shin
- White paper on "Active Magnetic Sensing for Long Duration Navigation Underwater," Office of Naval Research (PM: Tory Cobb) 2025
  Role: Wrote two research thrusts, "Active perception for robust map-based magnetic navigation" and "Active navigation with magnetic landmarks for map-free navigation". PI: Jaejeong Shin
- "Compliant mechanical Network Theory for Enhanced Design and Control of Robotic Systems," NSF Foundational Research in Robotics (FRR) 2023
  Role: Wrote the background, methodology, and experimental design sections. PI: Michael Griffis; Co-PIs: Carl D. Crane III, Shannon Ridgeway

PRESENTATIONS

- "Neural Network-Based Magnetometer Sensor Model for Virtual Magnetic-Based Navigation Testbed," AIAA SciTech Forum, Orlando, FL, January 2025 [C6]
- "Information-Driven Magnetic Navigation," research presentation and live demonstration, with an underwater motion-capture tank, for the Office of Naval Research program manager (Tory Cobb), University of Florida, Gainesville, FL, July 2024
- "Modularis: Modular Underwater Robot for Rapid Development and Validation of Autonomous Systems," poster presenting proposed Modularis 2.0 extensions to the platform of Herrin et al. (OCEANS 2023), UF Center for Coastal Solutions Summit, Gainesville, FL, December 2023

HONORS AND AWARDS

- Dean's Research Award, Herbert Wertheim College of Engineering, University of Florida 2023
  Competitive award funding Ph.D. studies.
- Achievement Award, Herbert Wertheim College of Engineering, University of Florida 2021
  Awarded on admission to the M.S. program.

PROFESSIONAL SERVICE

- Reviewer: The 26th International Conference on Autonomous Agents and Multiagent Systems (2026), IEEE International Conference on Robotics & Automation (2026-), IEEE/RSJ International Conference on Intelligent Robots and Systems (2025-), IEEE Transactions on Automation Science and Engineering (2026-), IEEE Journal of Oceanic Engineering (2026-)
- Member: IEEE Student Member, IEEE RAS Student Member, AIAA Student Member
- FIRST Robotics Mentor at P.K. Yonge High School: Mentored students on robotic design and software development
- First-Year Engineering 2025 Presentation on the Department of Mechanical and Aerospace Engineering